

SIMATS ENGINEERING

ASSESSMENT TOOL 3 - LOW (INDUSTRY PROBLEM SOLVING TASKS)

COURSE CODE: CSA09 COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO4: *Design and develop GUI-based user interfaces using Abstract Window Toolkit, Swing, and Applets by implementing event handling, layout managers, AWT controls, and menus. (BL3, BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 5 Total Marks: 50

ANNEXURE A INDUSTRY PROBLEM-SOLVING TASKS

Code of Conduct:

I MOHAMMAD IMRAN/192525329 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation:

Criteria	Understanding of the Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analyzing	Professionalism	Total (25%)	Total (100)
Weightage	20%	25%	25%	15%	10%	5%		
Q1								
Q2								
Q3								
Q4								

Q5								
Total								

ANNEXURE

INDUSTRY PROBLEM SOLVING TASKS

1. Hospital Patient Billing System

Develop a Java program for a hospital billing system using Class and Objects. Data Members:

- Patient ID
- Patient Name
- Room Type (General/Special)
- Number of Days Admitted
- Treatment Charges

Methods:

1. Read patient details.
2. Calculate room charges based on room type.
3. Calculate the total hospital bill.
4. Display patient billing details.

Note: General Room = ₹1,500/day, Special Room = ₹3,000/day.

2. Airline Ticket Reservation System

Develop a Java program for an airline reservation system using Class and Objects.

Data Members:

- Passenger ID
- Passenger Name
- Flight Number
- Ticket Fare
- Number of Tickets

Methods:

1. Read passenger details.
2. Calculate total ticket cost.
3. Apply a 5% discount if more than 5 tickets are booked.
4. Display booking and payment details.

Note: GST = 12% on the final amount.

3. Vehicle Rental Management System

Develop a Java program for a vehicle rental company using Class and Objects.

Data Members:

- Customer ID
- Customer Name
- Vehicle Type (Car/Bike)
- Number of Rental Days

Methods:

1. Read customer details.

2. Calculate rental charges.
3. Add a late return penalty if applicable.
4. Display the final rental bill.

Note: Car = ₹2,000/day, Bike = ₹500/day.

4. Manufacturing Production Monitoring System

Develop a Java program to monitor daily production in a manufacturing company using Class and Objects.

Data Members:

- Product ID
- Product Name
- Target Quantity
- Produced Quantity

Methods:

1. Read product details.
2. Compare actual production with the target.
3. Calculate production efficiency percentage.
4. Display production status (Achieved/Not Achieved).

Note: Efficiency = (Produced Quantity / Target Quantity) × 100.

5. Online Course Enrollment System

Develop a Java program for an online learning platform using Class and Objects.

Data Members:

- Student ID
- Student Name
- Course Name
- Course Fee

Methods:

1. Read student and course details.
2. Calculate the payable amount.
3. Provide a 15% scholarship discount if the fee exceeds ₹20,000.
4. Display enrollment and payment details.

Note: Registration fee = ₹500.

ANSWERS:

1. Hospital Patient Billing System

```
import java.util.Scanner;
```

```
class Patient {
    int patientId;
    String patientName;
    String roomType;
    int daysAdmitted;
    double treatmentCharges;

    void readDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Patient ID: ");
        patientId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Patient Name: ");
        patientName = sc.nextLine();

        System.out.print("Enter Room Type (General/Special): ");
        roomType = sc.nextLine();

        System.out.print("Enter Number of Days Admitted: ");
        daysAdmitted = sc.nextInt();

        System.out.print("Enter Treatment Charges: ");
        treatmentCharges = sc.nextDouble();
    }

    double calculateRoomCharges() {
        if (roomType.equalsIgnoreCase("General"))
            return daysAdmitted * 1500;
        else
            return daysAdmitted * 3000;
    }

    double calculateTotalBill() {
        return calculateRoomCharges() + treatmentCharges;
    }

    void displayDetails() {
        System.out.println("\nPatient ID: " + patientId);
        System.out.println("Patient Name: " + patientName);
        System.out.println("Room Charges: ₹" + calculateRoomCharges());
    }
}
```

```

        System.out.println("Treatment Charges: ₹" + treatmentCharges);
        System.out.println("Total Bill: ₹" + calculateTotalBill());
    }

    public static void main(String[] args) {
        Patient p = new Patient();
        p.readDetails();
        p.displayDetails();
    }
}

```

2. Airline Ticket Reservation System

```

import java.util.Scanner;

class AirlineReservation {
    int passengerId;
    String passengerName;
    String flightNumber;
    double ticketFare;
    int numberOfTickets;

    void readDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Passenger ID: ");
        passengerId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Passenger Name: ");
        passengerName = sc.nextLine();

        System.out.print("Enter Flight Number: ");
        flightNumber = sc.nextLine();

        System.out.print("Enter Ticket Fare: ");
        ticketFare = sc.nextDouble();

        System.out.print("Enter Number of Tickets: ");
        numberOfTickets = sc.nextInt();
    }

    double calculateTotalCost() {
        double total = ticketFare * numberOfTickets;

        if (numberOfTickets > 5)
            total = total - (total * 0.05);
    }
}

```

```

        double gst = total * 0.12;
        return total + gst;
    }

    void displayDetails() {
        System.out.println("\nPassenger ID: " + passengerId);
        System.out.println("Passenger Name: " + passengerName);
        System.out.println("Flight Number: " + flightNumber);
        System.out.println("Total Amount (with GST): ₹" + calculateTotalCost());
    }

    public static void main(String[] args) {
        AirlineReservation a = new AirlineReservation();
        a.readDetails();
        a.displayDetails();
    }
}

```

3. Vehicle Rental Management System

```

import java.util.Scanner;

class VehicleRental {
    int customerId;
    String customerName;
    String vehicleType;
    int rentalDays;
    double penalty;

    void readDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Customer ID: ");
        customerId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Customer Name: ");
        customerName = sc.nextLine();

        System.out.print("Enter Vehicle Type (Car/Bike): ");
        vehicleType = sc.nextLine();

        System.out.print("Enter Rental Days: ");
        rentalDays = sc.nextInt();

        System.out.print("Enter Late Return Penalty: ");
    }
}

```

```

        penalty = sc.nextDouble();
    }

    double calculateRentalCharges() {
        if (vehicleType.equalsIgnoreCase("Car"))
            return rentalDays * 2000;
        else
            return rentalDays * 500;
    }

    double finalBill() {
        return calculateRentalCharges() + penalty;
    }

    void displayDetails() {
        System.out.println("\nCustomer ID: " + customerId);
        System.out.println("Customer Name: " + customerName);
        System.out.println("Rental Charges: ₹" + calculateRentalCharges());
        System.out.println("Penalty: ₹" + penalty);
        System.out.println("Final Bill: ₹" + finalBill());
    }

    public static void main(String[] args) {
        VehicleRental v = new VehicleRental();
        v.readDetails();
        v.displayDetails();
    }
}

```

4. Manufacturing Production Monitoring System

```
import java.util.Scanner;
```

```

class Production {
    int productId;
    String productName;
    int targetQuantity;
    int producedQuantity;

    void readDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Product ID: ");
        productId = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Product Name: ");
    }
}

```

```

        productName = sc.nextLine();

        System.out.print("Enter Target Quantity: ");
        targetQuantity = sc.nextInt();

        System.out.print("Enter Produced Quantity: ");
        producedQuantity = sc.nextInt();
    }

    double calculateEfficiency() {
        return ((double) producedQuantity / targetQuantity) * 100;
    }

    void displayStatus() {
        System.out.println("\nProduct ID: " + productId);
        System.out.println("Product Name: " + productName);
        System.out.println("Efficiency: " + calculateEfficiency() + "%");

        if (producedQuantity >= targetQuantity)
            System.out.println("Status: Achieved");
        else
            System.out.println("Status: Not Achieved");
    }

    public static void main(String[] args) {
        Production p = new Production();
        p.readDetails();
        p.displayStatus();
    }
}

```

5. Online Course Enrollment System

```

import java.util.Scanner;

class CourseEnrollment {
    int studentId;
    String studentName;
    String courseName;
    double courseFee;

    void readDetails() {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student ID: ");
        studentId = sc.nextInt();
        sc.nextLine();
    }
}

```



```

        System.out.print("Enter Student Name: ");
        studentName = sc.nextLine();

        System.out.print("Enter Course Name: ");
        courseName = sc.nextLine();

        System.out.print("Enter Course Fee: ");
        courseFee = sc.nextDouble();
    }

    double calculatePayableAmount() {
        double fee = courseFee;

        if (courseFee > 20000)
            fee = fee - (courseFee * 0.15);

        return fee + 500; // Registration Fee
    }

    void displayDetails() {
        System.out.println("\nStudent ID: " + studentId);
        System.out.println("Student Name: " + studentName);
        System.out.println("Course Name: " + courseName);
        System.out.println("Payable Amount: ₹" + calculatePayableAmount());
    }

    public static void main(String[] args) {
        CourseEnrollment c = new CourseEnrollment();
        c.readDetails();
        c.displayDetails();
    }
}

```

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation